

SAFETY DATA SHEET

(SDS)

This safety data sheet complies with the requirements of:
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 2015/830

Revision Date 26-Dec-2023

WAI2 - EGHS - EUROPEAN

Revision Number 16

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product Identifier

Product Name pH 7.00 Buffer

Product No 910107

Unique Formula Identifier (UFI) Not applicable

REACH registration number Not applicable

Pure substance/mixture Mixture

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Manufacturer, Importer, Supplier Thermo Fisher Scientific©
Water and Lab Products
22 Alpha Road
Chelmsford, MA 01824, USA
1-978-232-6000

E-mail address wlp.techsupport@thermofisher.com

Made in USA

1.4. Emergency telephone number 24 Hour Emergency Phone Number
CHEMTREC®
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: 1-703-527-3887
(collect calls accepted)

SECTION 2. HAZARDS IDENTIFICATION**2.1. Classification of the substance or mixture**

Classification - Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Skin Corrosion/Irritation	Category 2
Serious eye damage/eye irritation	Category 2
Specific target organ toxicity (repeated exposure)	Category 2
Chronic aquatic toxicity	Category 3

2.2. Label elements**Signal Word**

Warning

Hazard Statements

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H373 - May cause damage to organs through prolonged or repeated exposure

H412 - Harmful to aquatic life with long lasting effects

EUH210 - Safety data sheet available on request

Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P273 - Avoid release to the environment

P264 - Wash face, hands and any exposed skin thoroughly after handling

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P314 - Get medical advice/attention if you feel unwell

P332 + P313 - If skin irritation occurs: Get medical advice/attention

P337 + P313 - If eye irritation persists: Get medical advice/attention

P362 - Take off contaminated clothing and wash before reuse

P501 - Dispose of contents/ container to an approved waste disposal plant

2.3. Other hazards**General Hazards**

This product does not contain any known or suspected endocrine disruptors
Toxic to terrestrial vertebrates

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component	EC No	CAS No	Weight %	CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567	REACH Reg. No
Water	EEC No. 231-791-2	7732-18-5	90 - 100%	Not classified	No information available
Sodium Phosphate, Dibasic	EEC No. 231-448-7	7558-79-4	0 - 10%	Not classified	No information available
Potassium Dihydrogen Phosphate	EEC No. 231-913-4	7778-77-0	0 - 10%	Not classified	No information available
5-Bromo-5-Nitro-1,3-Dioxane	EEC No. 250-001-7	30007-47-7	0.1-1%	Acute Tox. 4 - H302 Skin Corr. 1A - H314 Eye Damage 1 - H318 STOT Rep. Exp. 2 - H373 Aquatic Acute 1 - H400 Aquatic Chronic 1 - H410	No information available
FD & C Yellow #5	EEC No. 217-699-5	1934-21-0	<0.1%		No information available

Component	CAS No	Specific concentration limits (SCL's)	M-Factor	Component notes
Water	7732-18-5	-	-	-
Sodium Phosphate, Dibasic	7558-79-4	-	-	-
Potassium Dihydrogen Phosphate	7778-77-0	-	-	-
5-Bromo-5-Nitro-1,3-Dioxane	30007-47-7	-	-	-
FD & C Yellow #5	1934-21-0	-	-	-

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	Use first aid treatment according to the nature of the injury. For further assistance, contact your local Poison Control Center. Show this safety data sheet to the doctor in attendance.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.
Inhalation	Remove to fresh air. Get medical attention immediately if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.
Self-Protection of the First Aider	No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

Most important symptoms and effects	None reasonably foreseeable
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4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Personal Precautions	Ensure adequate ventilation. Use personal protective equipment as required.
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6.2. Environmental precautions

Environmental Precautions	Should not be released into the environment. See Section 12 for additional Ecological Information. Vapors may accumulate to form explosive concentrations.
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6.3. Methods and material for containment and cleaning up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

Reference to Other Sections

Refer to protective measures listed in Sections 7 and 8

See Section 8 for information on appropriate personal protective equipment

See Section 12 for additional Ecological Information

See Section 13 for additional waste treatment information

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Advice on safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Store at room temperature in the original container. Protect from direct sunlight.

7.3. Specific end use(s)

Specific Use(s)

Use as laboratory reagent

Risk Management Methods (RMM)

The information required is contained in this Safety Data Sheet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s):

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Sodium Phosphate, Dibasic	MAC: 10 mg/m ³				
Potassium Dihydrogen Phosphate	MAC: 10 mg/m ³				
5-Bromo-5-Nitro-1,3-Dioxane	Skin notation MAC: 3 mg/m ³				
FD & C Yellow #5	MAC: 5 mg/m ³				

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
5-Bromo-5-Nitro-1,3-Dioxane 30007-47-7 (0.1-1%)		DNEL = 7.78µg/kg bw/day		DNEL = 7.78µg/kg bw/day
FD & C Yellow #5 1934-21-0 (<0.1%)				DNEL = 52.82mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Potassium Dihydrogen Phosphate 7778-77-0 (0 - 10%)				DNEL = 14.82mg/m ³
5-Bromo-5-Nitro-1,3-Dioxane 30007-47-7 (0.1-1%)		DNEL = 27.4µg/m ³		DNEL = 27.4µg/m ³
FD & C Yellow #5 1934-21-0 (<0.1%)				DNEL = 372.52mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Sodium Phosphate, Dibasic 7558-79-4 (0 - 10%)	PNEC = 0.05mg/L		PNEC = 0.5mg/L	PNEC = 50mg/L	
FD & C Yellow #5 1934-21-0 (<0.1%)	PNEC = 0.12mg/L	PNEC = 0.46992mg/kg sediment dw	PNEC = 1.2mg/L	PNEC = 10mg/L	PNEC = 0.02353mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water intermittent	Food chain	Air
Sodium Phosphate, Dibasic 7558-79-4 (0 - 10%)	PNEC = 0.005mg/L				
FD & C Yellow #5 1934-21-0 (<0.1%)	PNEC = 0.012mg/L	PNEC = 0.046992mg/kg sediment dw			

8.2. Exposure controls**Engineering Measures**

None under normal use conditions

Personal protective equipment**Eye/face Protection**

Wear chemical splash goggles and face shield. If splashes are likely to occur: Goggles.

Skin and body protection Wear protective gloves/protective clothing.

Respiratory Protection No protective equipment is needed under normal use conditions.
Recommended Filter type: Particle filter.

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Light yellow
Odor	None
Odor Threshold	No information available
pH	7.00
PH Range	5.8 - 8.2

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling Point/Range	~ 100 °C / 212 °F	
Flash Point (High in °C)	No information available	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

9.2. Other information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

No information available

10.2. Chemical stability

Stable under normal conditions

Explosion Data

Sensitivity to Mechanical Impact	None
Sensitivity to Static Discharge	None

10.3. Possibility of hazardous reactions

None under normal processing

10.4. Conditions to avoid

Extremes of temperature and direct sunlight

10.5. Incompatible materials

No information available

10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors

SECTION 11: TOXICOLOGICAL INFORMATION**11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008****Product Information**

Acute Toxicity No information available
Unknown Acute Toxicity 0 % of the mixture consists of ingredient(s) of unknown toxicity.
The following values are calculated based on chapter 3.1 of the GHS document
ATEmix (oral) 38,924.00 mg/kg

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	LD50 > 90 mL/kg (Rat)		
Sodium Phosphate, Dibasic	LD50 = 17 g/kg (Rat)		
Potassium Dihydrogen Phosphate	LD50 = 3200 mg/kg (Rat)		LC50 > 0.83 mg/L (Rat) 4 h
5-Bromo-5-Nitro-1,3-Dioxane	LD50 = 455 mg/kg (Rat)		
FD & C Yellow #5	LD50 > 2000 mg/kg (Rat)		

Skin Corrosion/Irritation Irritating to skin
Serious eye damage/eye irritation Irritating to eyes
Sensitization No information available
Mutagenic Effects No information available
Carcinogenic effects No information available
Reproductive Effects No information available
STOT - single exposure No information available
STOT - repeated exposure Causes damage to organs through prolonged or repeated exposure
Target Organs None known.
Aspiration hazard No information available

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12. ECOLOGICAL INFORMATION**12.1. Toxicity****Ecotoxicity effects**

3.01% of the mixture consists of components(s) of unknown hazards to the aquatic environment

12.2. Persistence and degradability**Persistence**

Soluble in water, Persistence is unlikely, based on information available.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
5-Bromo-5-Nitro-1,3-Dioxane	1.6	No data available
FD & C Yellow #5	-1.572	No data available

12.4. Mobility in soil

Highly mobile in soils

Mobility

Will likely be mobile in the environment due to its water solubility

Component	log Pow
5-Bromo-5-Nitro-1,3-Dioxane 30007-47-7 (0.1-1%)	1.6
FD & C Yellow #5 1934-21-0 (<0.1%)	-1.572

12.5. Results of PBT and vPvB assessment

No information available

12.6. Endocrine disrupting properties

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

Ozone Depletion Potential

This product does not contain any known or suspected substance

SECTION 13. DISPOSAL CONSIDERATIONS**13.1. Waste treatment methods****Waste from Residues/Unused Products**

Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging

Improper disposal or reuse of this container may be dangerous and illegal.

SECTION 14: TRANSPORT INFORMATION**IMDG/IMO****14.1 UN-No**

Not Regulated

14.2 Proper Shipping Name

Not Regulated

14.3 Hazard Class

Not Regulated

14.4 Packing Group

Not Regulated

14.5 Marine Pollutant Not Applicable
 14.6 Special Provisions None
 14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code No information available

ADR

14.1. UN number Not Regulated
 14.2. UN proper shipping name Not Regulated
 14.3. Transport hazard class(es) Not Regulated
 14.4. Packing group Not Regulated

ICAO

14.1 UN-No Not Regulated
 14.2 Proper Shipping Name Not Regulated
 14.3 Hazard Class Not Regulated
 14.4 Packing Group Not Regulated
 14.5 Environmental hazard Not Applicable
 14.6 Special Provisions None

IATA

14.1 UN-No Not Regulated
 14.2 Proper Shipping Name Not Regulated
 14.3 Hazard Class Not Regulated
 14.4 Packing Group Not Regulated
 14.5 Environmental hazard Not Applicable
 14.6 Special Provisions None

SECTION 15: REGULATORY INFORMATION**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****International Inventories**

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS), U.S.A. (TSCA).

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Water	7732-18-5	231-791-2	-	-	X	X	KE-35400	X	-
Sodium Phosphate, Dibasic	7558-79-4	231-448-7	-	-	X	X	KE-12344	X	X
Potassium Dihydrogen Phosphate	7778-77-0	231-913-4	-	-	X	X	KE-28622	X	X
5-Bromo-5-Nitro-1,3-Dioxane	30007-47-7	250-001-7	-	-	X	X	KE-03687	-	-
FD & C Yellow #5	1934-21-0	217-699-5	-	-	X	X	KE-06857	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Water	7732-18-5	X	ACTIVE	X	-	X	X	X
Sodium Phosphate, Dibasic	7558-79-4	X	ACTIVE	X	-	X	X	X
Potassium Dihydrogen Phosphate	7778-77-0	X	ACTIVE	X	-	X	X	X
5-Bromo-5-Nitro-1,3-Dioxane	30007-47-7	X	ACTIVE	X	-	X	X	X
FD & C Yellow #5	1934-21-0	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

European Union

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Water	7732-18-5	-	-	-
Sodium Phosphate, Dibasic	7558-79-4	-	-	-
Potassium Dihydrogen Phosphate	7778-77-0	-	-	-
5-Bromo-5-Nitro-1,3-Dioxane	30007-47-7	-	-	-
FD & C Yellow #5	1934-21-0	-	Use restricted. See item 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

Water endangering class = 3 (self classification)

Component	Germany - Water Classification (AwSV)
Sodium Phosphate, Dibasic 7558-79-4 (0 - 10%)	WGK1
Potassium Dihydrogen Phosphate 7778-77-0 (0 - 10%)	WGK1
5-Bromo-5-Nitro-1,3-Dioxane 30007-47-7 (0.1-1%)	WGK3
FD & C Yellow #5 1934-21-0 (<0.1%)	WGK1

15.2. Chemical safety assessment

A Chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

SECTION 16: OTHER INFORMATION**Full text of H-Statements referred to under sections 2 and 3**

H302 - Harmful if swallowed

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

Key or legend to abbreviations and acronyms used in the safety data sheet**CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**TWA** TWA (time-weighted average)**Ceiling** Maximum limit value**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**STEL** STEL (Short Term Exposure Limit)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Full text of H-Statements referred to under section 3

H302 - Harmful if swallowed

H315 - Causes skin irritation

H373 - May cause damage to organs through prolonged or repeated exposure

H412 - Harmful to aquatic life with long lasting effects

Prepared By Regulatory Affairs**Prepared For** Thermo Fisher Scientific Inc.**Issue Date** No information available**Revision Date** 26-Dec-2023**Reason for revision** SDS sections updated.**Training Advice** Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.****Disclaimer**

IMPORTANT: The information contained in this SDS is correct to the best of our knowledge as of the issue date (or subsequent revision date, if any), and is to be used only as a guide. This SDS does not constitute a guarantee (express or implied) of any kind and we make no warranties of any kind as to the accuracy or completeness of the information contained herein or the merchantability or fitness of the product or this information for a particular purpose. It is the responsibility of each individual buyer/user to determine the suitability of this information and the product for its intended purposes. Product sales are subject to

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End of Safety Data Sheet